Question 1 – One Big Table (OBT)

Business Requirement

The Head of Retail Banking wants to identify the customers with the highest total account balances.

Step 1: Understand the KPI

KPI

Total Customer Balance

Formula

SUM(balance)
GROUP BY customer

Questions to Ask

- What is being measured?
- At what level is it measured?
- Who will consume the report?

Step 2: Identify Required Data

What data is needed to calculate the KPI?

Customer

customer_id
customer_name
city

Account

account_id
customer_id

balance

Step 3: Design the OBT

Grain

One row per account

Design

customer_balance_obt
---
account_id
customer_id
customer_name
city
balance

Step 4: Create Tables

Students should write:

CREATE TABLE customer_balance_obt (...)

Step 5: Write Business Query

```
SELECT
  customer_name,
  SUM(balance) AS total_balance
FROM customer_balance_obt
GROUP BY customer_name
ORDER BY total_balance DESC;
```

Question 2 – Kimball

Business Requirement

Management wants to analyze total deposits by branch.

Step 1: Understand the KPI

KPI

Total Deposits per Branch

Formula

SUM(balance)
GROUP BY branch

Step 2: Identify Required Data

Customer

customer_id
customer_name

Branch

branch_id
branch_name

Account

account_id
balance

Step 3: Design the Star Schema

Grain

One row per account balance

Fact Table

fact_account_balance

Measures:

balance

Dimensions

dim_customer
dim_branch

Step 4: Create Tables

Students should create:

```
CREATE TABLE dim_customer (...);

CREATE TABLE dim_branch (...);

CREATE TABLE fact_account_balance (...);
```

Step 5: Write Business Query

```
SELECT
  b.branch_name,
  SUM(f.balance) AS total_deposits
FROM fact_account_balance f
JOIN dim_branch b
ON f.branch_id = b.branch_id
GROUP BY b.branch_name;
```

Question 3 – Inmon

Business Requirement

Management wants a Customer 360 view including deposits and loans.

Step 1: Understand the KPI

KPI

Customer Value

Formula

Deposits + Loans

Step 2: Identify Required Data

Customer
Account
Loan
Branch

Step 3: Design 3NF Model

Identify entities:

Customer
Account
Loan
Branch

Identify relationships:

Customer → Account

Customer → Loan

Branch → Account

Normalize to 3NF.

Step 4: Create Tables

Students should create:

dim_customer

dim_branch

fact_account_balance

fact_loan

Step 5: Write Business Query

```
SELECT
  c.customer_name,
  SUM(ab.balance)
  +
  COALESCE(SUM(l.loan_amount),0)
  AS customer_value
FROM dim_customer c
LEFT JOIN fact_account_balance ab
ON c.customer_id = ab.customer_id
LEFT JOIN fact_loan l
ON c.customer_id = l.customer_id
GROUP BY c.customer_name;
```

Question 4 – Data Vault

Business Requirement

The bank receives customer, account, and loan data from multiple source systems.

Management wants a scalable architecture that supports future source systems.

Step 1: Understand the KPI

KPI

Customer Exposure

Formula

Deposits + Loans

Step 2: Identify Business Keys

customer_id

account_id

loan_id

Step 3: Design Data Vault

Hubs

hub_customer

hub_account

hub_loan

Links

link_customer_account

link_customer_loan

Satellites

sat_customer

sat_account

sat_loan

Step 4: Create Tables

Students should create:

hub_customer

hub_account

hub_loan

link_customer_account

link_customer_loan

sat_customer

sat_account

sat_loan

Step 5: Write Business Query

Create a business view first:

customer_360_view

Then answer:

Which customers have the highest exposure?

Example:

```
SELECT
  customer_id,
  SUM(balance)
  +
  SUM(loan_amount)
FROM customer_360_view
GROUP BY customer_id;
```

This structure teaches students to think like a data architect:

Business Requirement

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Understand KPI

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Identify Required Data

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Choose Modeling Approach

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Design Tables

↓

Load Data

↓

Answer Business Question